Roadmap Master Java Multithreading & Concurrency

📚 SÁCH KINH ĐIỂN (Must Read)

1. ⭐⭐⭐⭐⭐ "Java Concurrency in Practice" - Brian Goetz

THE BIBLE của Java Concurrency

Tác giả: Brian Goetz (Lead architect của Java's concurrency libraries)

Nội dung:

- Phần I: Fundamentals
  - Thread safety
  - Sharing objects
  - Composing objects
- Phần II: Structuring Concurrent Applications
  - Task execution
  - Cancellation and shutdown
  - Thread pools
- Phần III: Liveness, Performance, and Testing
  - Avoiding liveness hazards
  - Performance and scalability
  - Testing concurrent programs
- Phần IV: Advanced Topics
  - Explicit locks
  - Building custom synchronizers
  - Atomic variables and nonblocking synchronization

Tại sao phải đọc: ✅ Được viết bởi người thiết kế Java concurrency APIs ✅ Giải thích CHI TIẾT về memory model, happens-before ✅ Best practices từ thực tế production ✅ Covers java.util.concurrent package từ trong ra ngoài ✅ Pattern và anti-pattern rõ ràng

Cấp độ: Intermediate → Advanced

Độ khó: ⭐⭐⭐⭐ (Khó nhưng rất đáng)

Link:

- Amazon: https://www.amazon.com/Java-Concurrency-Practice-Brian-Goetz/dp/0321349601

- PDF: (tìm trên Z-Library hoặc mua bản gốc)

Lộ trình đọc:

```
Week 1-2: Part I (Fundamentals)
  → Hiểu thread safety, synchronization, visibility

Week 3-4: Part II (Structuring Applications)
  → Executor framework, Thread pools (ĐÚNG VỚI VẤN ĐỀ CỦA BẠN!)

Week 5-6: Part III (Performance & Testing)
  → Deadlock, livelock, performance tuning

Week 7-8: Part IV (Advanced)
  → Lock, Atomic, CAS operations
```

Chapters quan trọng nhất cho bạn:

- **Chapter 6: Task Execution** ← ĐÚNG VỚI CODE CỦA BẠN
- **Chapter 7: Cancellation and Shutdown** ← Graceful shutdown
- **Chapter 8: Applying Thread Pools** ← ThreadPoolExecutor configuration
- **Chapter 10: Avoiding Liveness Hazards** ← Deadlock, starvation
- **Chapter 11: Performance and Scalability** ← Tuning thread pools

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2. ⭐⭐⭐⭐⭐ "Concurrent Programming in Java" - Doug Lea

Sách từ cha đẻ của java.util.concurrent

Tác giả: Doug Lea (Người tạo ra java.util.concurrent package!)

Nội dung:

- Concurrent Object-Oriented Programming
- Exclusion and synchronization
- State dependence
- Creating threads
- Before/after preconditions

Tại sao phải đọc: ✅ Được viết bởi người TẠO RA java.util.concurrent ✅ Explain design decisions đằng sau APIs ✅ Pattern-oriented approach ✅ Deep dive vào memory models

Cấp độ: Advanced

Độ khó: ⭐ ⭐ ⭐ ⭐ ⭐ (Rất khó, academic style)

Note: Sách này cũ hơn (1999) nhưng foundations vẫn đúng. Đọc SAU "Java Concurrency in Practice"

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3. ⭐ ⭐ ⭐ ⭐ "The Art of Multiprocessor Programming" - Maurice Herlihy & Nir Shavit

Nội dung:

- Mutual exclusion
- Concurrent objects
- Foundations of shared memory
- Futures, scheduling, work distribution
- Lock-free programming
- Transactional memory

Tại sao đọc: ✅ Không chỉ Java, cover toàn bộ multiprocessor theory ✅ Algorithms cho concurrent data structures ✅ Lock-free & wait-free programming ✅ Có cả exercises và solutions

Cấp độ: Advanced → Expert

Độ khó: ⭐ ⭐ ⭐ ⭐ ⭐ (Rất academic, nhiều toán)

Khi nào đọc: Khi bạn đã master "Java Concurrency in Practice"

📖 SÁCH BỔ TRỢ

4. ⭐ ⭐ ⭐ ⭐ "Effective Java" - Joshua Bloch (3rd Edition)

Chapters liên quan:

- Chapter 11: Concurrency (Items 78-84)
  - Item 78: Synchronize access to shared mutable data
  - Item 79: Avoid excessive synchronization
  - Item 80: Prefer executors, tasks, and streams to threads
  - Item 81: Prefer concurrency utilities to wait and notify
  - Item 82: Document thread safety
  - Item 83: Use lazy initialization judiciously
  - Item 84: Don't depend on the thread scheduler

Tại sao đọc: ✅ Best practices ngắn gọn, súc tích ✅ Item-based, dễ reference ✅ Modern Java (Java 9+) ✅ Real-world examples

Cấp độ: Intermediate

Độ khó: ⭐ ⭐ ⭐

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5. ⭐ ⭐ ⭐ ⭐ "Java Performance: The Definitive Guide" - Scott Oaks

Chapters liên quan:

- Chapter 9: Threading and Synchronization Performance
- Chapter 10: Java Servers

Nội dung:

- Thread pool sizing
- ForkJoinPool
- Thread synchronization costs
- Lock contention
- Profiling concurrent applications

Tại sao đọc: ✅ Focus vào PERFORMANCE ✅ How to measure và tune ✅ JVM internals cho concurrency ✅ Real profiling tools và techniques

Cấp độ: Intermediate → Advanced

Độ khó: ⭐ ⭐ ⭐ ⭐

🎥 VIDEO COURSES

1. ⭐ ⭐ ⭐ ⭐ ⭐ "Java Multithreading, Concurrency & Performance Optimization" - Udemy

Instructor: Michael Pogrebinsky

Link: https://www.udemy.com/course/java-multithreading-concurrency-performance-optimization/

Nội dung:

- Thread fundamentals
- Performance optimization
- Data sharing between threads

- Advanced locking
- Lock-free algorithms
- Thread pooling and executors

Tại sao tốt: ✅ Hands-on coding exercises ✅ Visual explanations ✅ Modern Java (Java 11+) ✅ Performance testing included ✅ 16+ hours content

Giá: ~$15-20 (khi sale)

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2. ⭐⭐⭐⭐ "Parallel and Concurrent Programming with Java" - LinkedIn Learning

Instructors: Olivia Chiu Stone & Barron Stone

Link: https://www.linkedin.com/learning/paths/parallel-and-concurrent-programming-with-java

Nội dung:

- Part 1: Threads and locks
- Part 2: Parallel algorithms
- Part 3: Concurrent data structures

Tại sao tốt: ✅ Structured path ✅ Good for beginners ✅ Visual demos ✅ Quizzes and exercises

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📝 ONLINE RESOURCES

1. ⭐⭐⭐⭐⭐ Oracle Java Tutorials - Concurrency

Link: https://docs.oracle.com/javase/tutorial/essential/concurrency/

Nội dung:

- Processes and Threads
- Thread Objects
- Synchronization
- Liveness
- Guarded Blocks
- Immutable Objects
- High Level Concurrency Objects

Tại sao tốt: ✅ OFFICIAL documentation ✅ Free ✅ Up-to-date ✅ Code examples

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2. ⭐⭐⭐⭐⭐ Baeldung - Java Concurrency

Link: https://www.baeldung.com/java-concurrency

Topics:

- Thread Pools
- ExecutorService
- CompletableFuture
- Synchronization
- Locks
- Concurrent Collections

Tại sao tốt: ✅ Practical examples ✅ Modern Java ✅ Copy-paste ready code ✅ Updated regularly

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3. ⭐⭐⭐⭐ Java Concurrency API Documentation

Link: https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/util/concurrent/package-summary.html

Đọc Javadoc của:

- ThreadPoolExecutor
- ExecutorService
- Future, CompletableFuture
- ConcurrentHashMap
- Atomic classes
- Locks (ReentrantLock, ReadWriteLock)

Tại sao quan trọng: ✅ Source of truth ✅ Implementation details ✅ Performance characteristics

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🎯 LỘ TRÌNH HỌC (RECOMMENDED PATH)

Phase 1: Foundations (2-3 tuần)

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1. Oracle Java Tutorials - Concurrency
   → Hiểu basics: Thread, Runnable, synchronized

2. Effective Java - Chapter 11
```

→ Best practices ngắn gọn

3. Baeldung tutorials
→ Hands-on với ExecutorService, Thread Pools

Phase 2: Deep Dive (2-3 tháng)

1. Java Concurrency in Practice (FULL BOOK)
→ Đọc kỹ từng chapter
→ Làm exercises
→ Apply vào code của bạn

Priority chapters:
- Chapter 6: Task Execution
- Chapter 7: Cancellation and Shutdown
- Chapter 8: Applying Thread Pools
- Chapter 13: Explicit Locks

Phase 3: Advanced (1-2 tháng)

1. Java Performance - Chapter 9-10
→ Performance tuning
→ Profiling

2. The Art of Multiprocessor Programming
→ Lock-free algorithms
→ Advanced concurrent data structures

Phase 4: Practice (Continuous)

1. Implement common patterns:
- Producer-Consumer
- Thread Pool từ đầu
- Lock-free queue
- Fork-Join tasks

2. Contribute to open source:
- Analyze code của libraries lớn
- Submit PRs fix concurrency issues

3. Build projects:
- Web crawler (multi-threaded)

- Download manager
- Chat server
- Task scheduler

💡 STUDY TIPS

1. Đọc code thực tế

java

```
// Đọc source code của JDK:
- java.util.concurrent.ThreadPoolExecutor
- java.util.concurrent.ForkJoinPool
- java.util.concurrent.ConcurrentHashMap
- java.util.concurrent.locks.ReentrantLock

// Clone từ GitHub:
git clone https://github.com/openjdk/jdk.git
cd jdk/src/java.base/share/classes/java/util/concurrent
```

2. Viết test cases

java

```
// Test thread safety:
@Test
public void testConcurrentAccess() throws InterruptedException {
    Counter counter = new Counter();
    int numThreads = 100;
    int incrementsPerThread = 1000;

    ExecutorService executor = Executors.newFixedThreadPool(numThreads);
    CountDownLatch latch = new CountDownLatch(numThreads);

    for (int i = 0; i < numThreads; i++) {
        executor.submit(() -> {
            for (int j = 0; j < incrementsPerThread; j++) {
                counter.increment();
            }
            latch.countDown();
        });
    }

    latch.await();
    assertEquals(numThreads * incrementsPerThread, counter.get());
}
```

3. Sử dụng tools

```
bash

# Thread dump analysis
jstack <pid> > thread_dump.txt

# Find deadlocks
jstack <pid> | grep -A 10 "deadlock"

# Profiling với JProfiler/YourKit
# Visual VM (free)
jvisualvm

# Java Flight Recorder
java -XX:StartFlightRecording ...
```

4. Tham gia cộng đồng

- StackOverflow - tag [java] [concurrency]

- Reddit - r/java
- Java Ranch forums
- Concurrency mailing lists

🎯 SPECIFIC TO YOUR CODE

Đọc ngay các phần này:

Từ "Java Concurrency in Practice":

Chapter 6: Task Execution

- Section 6.2: The Executor Framework ← YOUR CODE
- Section 6.3: Finding Exploitable Parallelism

Chapter 7: Cancellation and Shutdown

- Section 7.3: Handling Abnormal Thread Termination
- Section 7.4: JVM Shutdown

Chapter 8: Applying Thread Pools

- Section 8.1: Implicit Couplings Between Tasks and Execution Policies
- Section 8.2: Sizing Thread Pools ← QUAN TRỌNG!
- Section 8.3: Configuring ThreadPoolExecutor ← YOUR CODE
- Section 8.4: Extending ThreadPoolExecutor
- Section 8.5: Parallelizing Recursive Algorithms

Chapter 10: Avoiding Liveness Hazards

- Section 10.1: Deadlock
- Section 10.2: Avoiding and Diagnosing Deadlocks

Chapter 11: Performance and Scalability

- Section 11.4: Reducing Lock Contention
- Section 11.5: Example: Comparing Map Performance

📊 CHEAT SHEET - Key Concepts

1. Thread Safety Techniques

1. Immutability (best)
2. Synchronization (synchronized keyword)
3. Volatile variables
4. Atomic variables (AtomicInteger, etc.)
5. Concurrent collections (ConcurrentHashMap)
6. Explicit locks (ReentrantLock)
7. ThreadLocal

2. Executor Types

newFixedThreadPool(n) → Fixed size, unbounded queue ❌
newCachedThreadPool() → 0 to Integer.MAX, 60s timeout
newSingleThreadExecutor() → Single thread, unbounded queue
newScheduledThreadPool(n) → Scheduled/periodic tasks
newWorkStealingPool() → ForkJoinPool, work stealing

Custom ThreadPoolExecutor → Full control ✅ (YOUR CASE)

3. Common Pitfalls

❌ Unbounded queues → OOM
❌ No timeout on futures → Hung threads
❌ Wrong rejection policy → Lost data
❌ Excessive synchronization → Deadlock
❌ Not shutting down executors → Resource leak
❌ Shared mutable state without sync → Race conditions

🏆 MASTERY CHECKLIST

Level 1: Beginner ✅

- [ ] Understand Thread vs Runnable
- [ ] Know synchronized keyword
- [ ] Understand race conditions
- [ ] Can use ExecutorService
- [ ] Know basic concurrent collections

Level 2: Intermediate

- [ ] Understand memory model & happens-before
- [ ] Can configure ThreadPoolExecutor properly
- [ ] Know when to use different locks
- [ ] Can identify and fix deadlocks
- [ ] Understand volatile and atomic variables
- [ ] Can write thread-safe code

Level 3: Advanced

- [ ] Can design concurrent architectures
- [ ] Understand lock-free algorithms
- [ ] Can tune thread pools for performance
- [ ] Can analyze thread dumps
- [ ] Know internal implementation of concurrent collections
- [ ] Can write custom synchronizers

Level 4: Expert

- [ ] Contribute to concurrent libraries
- [ ] Can implement lock-free data structures
- [ ] Understand CPU cache coherence
- [ ] Can optimize for specific hardware
- [ ] Published papers/talks on concurrency
- [ ] Can teach others effectively

📚 SUMMARY - YOUR ACTION PLAN

Tuần 1-2:

✅ Đọc Oracle Java Tutorials - Concurrency
✅ Đọc Effective Java - Item 78-84
✅ Làm quen với Baeldung examples

Tháng 1-3:

📖 ĐỌC KỸ "Java Concurrency in Practice"
- Chapter 6, 7, 8 (priority cao)
- Chapter 10, 11

- Làm exercises
- Refactor code hiện tại theo best practices

Tháng 4-6:

📖 Đọc "Java Performance" - Chapter 9-10
🎥 Xem Udemy course để củng cố
💻 Build side projects:
- Multi-threaded web crawler
- Concurrent task scheduler
- Load testing framework

Continuous:

🔍 Đọc source code JDK concurrent package
🐛 Fix concurrency bugs trong open source
📝 Viết blog về những gì học được
👥 Tham gia discussions trên StackOverflow

⚡ QUICK WIN

Nếu chỉ có THỜI GIAN ĐỌC 1 QUYỂN:

→ **"Java Concurrency in Practice"** by Brian Goetz

Lý do:

- Được viết bởi lead architect
- Cover 90% những gì bạn cần
- Practical, không quá academic
- Best practices từ production systems
- Apply được ngay vào code của bạn

HOW TO READ:

1. Đọc lướt toàn bộ 1 lần (2 tuần)
2. Đọc kỹ chapters 6-8, 10-11 (2 tuần)
3. Làm exercises (1 tuần)
4. Refactor code cũ (1 tuần)

5. Reference khi cần

Thời gian: ~2 tháng để master

Chúc bạn thành công! 🚀